

Elementary Group Theory Notes

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1 Introduction

Reading from Dummit & Foote Abstract Algebra as well as professor Keith Conrad's notes. These notes are meant to be very brief and just small reminders of the bigger picture that can be found in Dummit & Foote's abstract algebra book. (Really they are just meant for me to skim for the sake of retention and understanding haha)

2 Symmetric Groups

Let Ω be any non-empty set and let S_Ω be a set of bijections from Ω to Ω . The set S_Ω is a group under function composition.

If $\sigma : \Omega \rightarrow \Omega$ and $\tau : \Omega \rightarrow \Omega$ are both bijections then: $\sigma \circ \tau$ is also a bijection for $\sigma \circ \tau : \Omega \rightarrow \Omega$. The identity of S_Ω is the permutation defined by $1(a) = a, \forall a \in \Omega$. For every permutation σ there is an inverse map $\sigma^{-1} : \Omega \rightarrow \Omega$ where $\sigma \circ \sigma^{-1} = \sigma^{-1} \circ \sigma = 1$. And $(S_\Omega, \circ)$ satisfies all other group axioms. We denote the language when talking about these groups "*Symmetric group on set Ω* ", where the elements of S_Ω are permutations of Ω .

We wish to show that: $|S_n| = n!$.

Proof. The permutations of $\{1, 2, 3, 4, \dots, n\}$ are the injective functions from this set to itself: $\sigma(1)$ sends 1 to any of n elements of $\{1, 2, 3, \dots, n\}$; $\sigma(2)$ can be any one of these elements except where $\sigma(1)$, notice that there are $(n - 1)$ choices for $\sigma(2)$. These process repeats and we notice that the number of permutations possible (the order of S_n is $n!$) $\square$

Writing elements σ of S_n involves *cycle decomposition*. A cycle is a string of integers which represents the elements of S_n - cyclically permutes these integers. The cycle $(a_1 a_2 \dots a_n)$ is the permutation that sends a_i to a_{i+1} , $1 \leq i \leq n - 1$ and sends a_n to a_1 .

Example: $(2, 1, 3)$ maps $2 \rightarrow 1$, $1 \rightarrow 3$, $3 \rightarrow 2$.

In general for each $\sigma \in S_n$, the numbers 1 through n are rearranged and grouped into k cycles of $(a_1 a_2 \dots a_{m_1}) (a_{m_1+1} a_{m_1+2} \dots a_{m_2}) \dots (a_{m_{k-1}+1} a_{m_{k-1}+2} \dots a_{m_k})$. Where the product of all cycles is the cycle decomposition of σ .

The Cycle Decomposition Algorithm: To start a new cycle, start with the smallest element and see where the element maps to until there is a repeat. Rinse and repeat this process with the rest of the elements within the cycle. Remove cycles of length 1. **COME BACK TO THIS LATER**

3 Matrix Groups

Before introducing Matrix Groups it is important to introduce the basic notion of a field. A Field is the smallest mathematical structure in which we can perform $+$, $-$, $\times$, $\div$, every non-zero element must have a multiplicative inverse.

Definition 1. A field is a set $\mathbb{F}$ with two binary operations $+$, $\cdot$ of $\mathbb{F}$ such that $(\mathbb{F}, +)$ is abelian with identity 0 and $(\mathbb{F} - \{0\}, \cdot)$ is also abelian. The following distribution laws hold:

$-a \cdot (b + c) = (a \cdot b) + (a \cdot c), \forall a, b, c \in \mathbb{F}$. Additionally for any field $\mathbb{F}$ let $\mathbb{F}^* = \mathbb{F} - \{0\}$.

Definition 2. For all positive integers $n \in \mathbb{Z}^+$. Let $GL_n(\mathbb{F})$ be the set of all $n \times n$ matrices whose entries are from $\mathbb{F}$ and $\det(A) \neq 0$.

$$GL_n(\mathbb{F}) = \{A | A = n \times n, \det(A) \neq 0\}$$

$GL_n(\mathbb{F})$ is closed under Matrix Multiplication. By the Invertible Matrix Theorem we also know that $\det(A) \neq 0$ if and only if A has an inverse. So each $A \in GL_n(\mathbb{F})$ has an inverse $A^{-1} \in GL_n(\mathbb{F})$, where $AA^{-1} = A^{-1}A = I_n$.

Definition 3. A special subgroup of $GL_n(\mathbb{F})$ is defined as $SL_n(\mathbb{F})$ which are matrices closed under matrix multiplication where $A \in SL_n(\mathbb{F}), \det(A) = 1$. The name *special* is kind of a funny word to use, but all of the same group laws hold.

4 Homomorphisms & Isomorphisms

Definition 4. Let $(G, *)$, (H, τ) be groups with some map $\phi : G \rightarrow H$ such that $\phi(x * y) = \phi(x)\tau\phi(y), \forall x, y \in G$. A map with this associated property is known as a Homomorphism.

It is important to not that the product on the left is computed in G and the product on the right is computed in H . A group homomorphism between two groups preserves the group operation but not necessarily uniqueness, as it is possible for multiple elements to map to one.

Definition 5. The map $\phi : G \rightarrow H$ is an isomorphism $G \cong H$ if. **1.)** ϕ is a homomorphism, and **2.)** ϕ is a bijection.

MENTION SIMILARITIES BETWEEN GROUPS THROUGH ISOMORPHISMS

Example 4.1. Take $\exp : \mathbb{R} \rightarrow \mathbb{R}^+$ where $\exp(x) = e^x$. We can say that this is an isomorphism from $(\mathbb{R}, +) \rightarrow (\mathbb{R}^+, \times)$, $\exp(y + x) = e^y e^x$ and $\exp(y) \cdot \exp(x) = e^x e^y$. Both of the elements and operations are different in $(\mathbb{R}, +)$ and $(\mathbb{R}^+, \times)$, yet the groups are isomorphic as groups they have identical structures.

A key idea from this example is an isomorphism preserving the structure from one group to another. We would not necessarily be able to tell the difference between two isomorphic structures if only given some abstract information about their elements and how they act upon each other. Take for example "Any non-abelian group of order 6 is isomorphic to S_3 ". Examples being: $D_6 \cong S_3$, and $GL_2(\mathbb{F}_2) \cong S_3$.

5 Basics of Subgroups

Definition 6. A subset H of a group G is a subgroup if H is non-empty and H is closed under products and inverses. We denote this as $H \leq G$. (**i.e** $x, y \in H \implies x^{-1} \in H$ **and** $xy \in H$). It is

important to notice that every subgroup is non-empty and must contain 1, the identity of G . The subgroup that only contains 1 is the *trivial subgroup*.

The subgroup criterion are the following: A subset H of G is a subgroup if and only if: **1.)** $H \neq \emptyset$ and **2.)** $\forall x, y \in H, xy^{-1} \in H$

Proof. Let $x \in H$, let $x = y$ we can deduce that $xx^{-1} = 1 \in H$, where H is non-empty because it contains the identity of G . Since H contains 1 then $1(x^{-1}) \in H$, i.e, $x^{-1} \in H$ and H is closed under taking inverses. If $x, y \in H$ where $x \neq y$, we know that $y^{-1} \in H$. H also contains $x(y^{-1})^{-1} = xy$, hence H is also closed under multiplication, which proves that H is also a subgroup of G . $\square$

6 Subgroups Generated by Subsets of a Group

Let G be any group and A be a subset of G . We will now study the Subgroup of G generated by A .

Proposition 7. IF A is any non-empty collection of Subgroups of G , then the intersection of all members of A is also a subgroup of G .

Proof. Let

$$K = \bigcap_{H \in A} H$$

Since each $H \in A$ is a subgroup then $1 \in H \implies 1 \in K$ This shows that $K \neq \emptyset$. Now lets consider $a, b \in K \implies a, b \in H, \forall H \in A$. Since each $H \in A$ is a group then $ab^{-1} \in H, \forall H$, hence $ab^{-1} \in K$. Therefore $K \leq G$. $\square$

Definition 8. If A is any subset of G , define

$$\langle A \rangle = \bigcap_{\substack{A \subseteq H \\ H \leq G}} H$$

To be the subgroup of G generated by A . $\langle A \rangle$ is the intersection of all subgroups of G containing A . $\mathcal{A} = \{H \leq G | A \subset H\}$.

Since A lies in each $H \in \mathcal{A}$, A is a subset of their intersection, $\langle A \rangle$. We denote $\langle A \rangle$ as the unique minimal element of $\mathcal{A}$. $\langle A \rangle$ is a subgroup of G containing A , so $\langle A \rangle \in \mathcal{A}$ and any element of $\mathcal{A}$ contains the intersection of all elements in $\mathcal{A}$ (i.e contains $\langle A \rangle$).

Proposition 9. Let

$$\bar{A} = \{a_1^{e_1} \cdot a_2^{e_2} \cdot a_3^{e_3} \cdot \dots a_n^{e_n} | n \in \mathbb{Z}, n \geq 0, \text{ and, } a_i \in A, e_i = \pm 1 \forall i \in \mathbb{Z}\}$$

Where $\bar{A} = \{1\}$ if $A = 0$. We call $\bar{A}$ the set of all finite products of elements of A and inverses of elements of A .

Proposition 10.

$$\bar{A} = \langle A \rangle$$

Proof. First we prove that $\bar{A}$ is a subgroup: $a, b \in \bar{A}$ where $a = a_1^{e_1} \cdot a_2^{e_2} \cdot a_3^{e_3} \cdot \dots a_n^{e_n}$ and $b = b_1^{\beta_1} \cdot b_2^{\beta_2} \cdot b_3^{\beta_3} \cdot \dots b_n^{\beta_n}$. The product of $(a \cdot b) = a_1^{e_1} \cdot a_2^{e_2} \cdot a_3^{e_3} \cdot \dots a_n^{e_n} \cdot b_1^{\beta_1} \cdot b_2^{\beta_2} \cdot b_3^{\beta_3} \cdot \dots b_n^{\beta_n}$. (ab) is a product of elements of A where $e_i, \beta_i = \pm 1$ therefore $(a \cdot b) \in \bar{A}$ (closed under multiplication).

Checking for inverses we find $b^{-1} = b_1^{-\beta_1} \cdot b_2^{-\beta_2} \cdot b_3^{-\beta_3} \cdot \dots \cdot b_n^{-\beta_n}$ where again we see $b^{-1} \in \bar{A}$. Therefore $\bar{A} \leq G$.

Since each $a \in A$ may be written a^1 , it follows that $A \subset \bar{A}$, hence $\langle A \rangle \subset \bar{A}$. We know that $\langle A \rangle$ is a group containing A and since it is closed under the group operation and taking inverses, $\langle A \rangle$ contains each element of the form $a_1^{e_1} \cdot a_2^{e_2} \cdot a_3^{e_3} \cdot \dots \cdot a_n^{e_n}$, meaning $\bar{A} \subset \langle A \rangle$. Therefore $\bar{A} = \langle A \rangle$. $\square$

We end the section with

$$\langle A \rangle = \{a_1^{e_1} \cdot a_2^{e_2} \cdot a_3^{e_3} \cdot \dots \cdot a_n^{e_n} \mid e_i \in \mathbb{Z}, \forall i\}$$

where each $(a_i^{e_i} \neq a_{i+1}^{e_{i+1}})$. Note that this product is taken group grouping like bases $a_i = a_i$ and consolidating/ combining exponents. Each a_i has finite order d_i therefore

$$|\langle A \rangle| \leq d_1 d_2 \dots d_k$$

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7 Quotient Groups & Homomorphisms

If ϕ is a group homomorphism from groups $\phi : G \rightarrow H$, the fibers of ϕ are sets of elements of G projecting to single elements of H . Formally:

Definition 11. Given some map $f : X \rightarrow Y$ A fiber is the pre-image of a singleton set $\{y\}$ where $y \in Y$, that is

$$f^{-1}(y) = \{x : f(x) = y\}$$

is the fiber of f over y

$$f^{-1}(y) = f^{-1}(\{y\})$$

The group G can now effectively be seen as partitioned by these fibers, where the fibers make up the structure of the Quotient Group. However, there is more that must be introduced first before talking about this Quotient Group.

Proposition 12. The set of fibers is a group:

If X_a is a fiber above a , and X_b is a fiber above b , then $X_a \cdot X_b = X_{ab}$. This multiplication is associative. The inverse of a fiber of a is the fiber associated with a^{-1} .

We will now introduce the idea of a Kernel which is essential for the understanding of Quotient Groups:

Definition 13. If ϕ is a group homomorphism from $\phi : G \rightarrow H$, the **kernel** of ϕ is the set:

$$\{g \in G : \phi(g) = 1\} = \ker(\phi)$$

In plain language the Kernel of ϕ is the set of elements in G that map to the singular identity element of H .

Example 7.1. Take

$$\phi(gg^{-1}) = \phi(g) \cdot \phi(g^{-1}) = 1 \implies \phi(g^{-1}) = \frac{1}{\phi(g)}$$

We just found that

$$\phi(g^{-1}) = \phi(g)^{-1}$$

There are other additional properties regarding kernels that are worth mentioning but I will not necessarily prove in these notes:

- $\phi(1_G) = \phi(1_H)$
- $\phi(g^{-1}) = \phi(g)^{-1}$
- $\phi(g^n) = \phi(g)^n, \forall n \in \mathbb{Z}$
- $\ker(\phi)$ is a subgroup of G
- $\text{img}(\phi)$, image of G under ϕ is a subgroup of H

We now have the necessary language to introduce the basic definition of a Quotient Group

Definition 14. Let $\phi : G \rightarrow H$ be a homomorphism with kernel K . The **Quotient Group** G/K , is a group whose elements are fibers of ϕ with group operation defined above.

This definition gives us some insight into the structure of the Quotient Group as the kernel K is almost "dividing out" or collapsing the larger group G . Essentially what is happening is we take a larger group and collapse it into an identically structured smaller group with the elements partitioned into fibers of the larger group.

Proposition 15. Let $\phi : G \rightarrow H$ be a homomorphism with $K = \ker(\phi)$. Let $X \in G/K$ be a fiber above a (i.e $X = \phi^{-1}(a)$). Then:

- $\forall u \in X : X = \{uk : k \in K\}$
- $\forall u \in X, X = \{ku : k \in K\}$

The sets arising in this proposition essentially tell us that any element (fiber) that belongs to a quotient group can be constructed by a single element (the representative of that fiber) and the elements of the kernel K .

Proof. We will prove just the first property but the second follows similarly: Let $u \in X$ so by definition $\phi(u) = a$. Let

$$uK = \{uk : k \in K\}$$

First we prove that $uK \subseteq X$ for any $k \in K$: Take $\phi(uk) = \phi(u)\phi(k) = a\phi(k) = a(1)$ (Since k belongs to K) $= a$. We just showed that $uk \in X$ for any k , therefore $uK \subseteq X$. Now suppose that $g \in X$ and let $k = u^{-1}g$. We find that $\phi(k) = \phi(u^{-1})\phi(g) = \phi(u)^{-1}\phi(g) = a^{-1}a = 1$ (identity element). We just showed that $k \in \ker(\phi)$. Since $g = uk \in uK$ we just established $X \subseteq uK$. Finishing the proof. $\square$

These types of sets that arise in proposition 2 do not necessarily need to be kernels.

Definition 16. For any $N \leq G$ and any $g \in G$, Let

$$gN = \{gn : n \in N\} \text{ and, } Ng = \{ng : n \in N\}$$

be the left coset and right coset respectively of N in G .

Any element of a coset can be used as a singular representative of a coset. Multiplication in the quotient group can be defined in terms of representatives for the cosets. Fibers are cosets! We can also draw the conclusion that elements of a quotient group are left cosets $gK, g \in G$.

Example 7.2. Take for example $\mathbb{Z}/4\mathbb{Z}$. Each congruence class can be viewed as a coset. To spell this out more the fibers of $\mathbb{Z}/4\mathbb{Z}$ are $[1], [2], [3]$ where each of these elements multiplied by 4 form the coset.

To construct a group out of these cosets we introduce (but not yet prove) the following theorem:

Theorem 17. *Let G be a group, let K be the Kernel of the homomorphism $\phi : G \rightarrow H$. Then the set whose elements are the left cosets of K in G with operation*

$$uK \circ vK = (uv)K$$

forms a group, G/K where this operation is well defined.